Marketing budget for next quarter — what each channel returns and where the 18M should go

To: CMO, CFO · From: Yaswanth Krishna · Data: 40 regions × 156 weeks (Jan 2021 – Jan 2024) · Model trained to 16 Oct 2023 and checked on the 13 weeks after · Ranges are 90% intervals · Source: `reports/project_b/results.md`

Recommendation

Spend the 18M (13% less than last quarter's 20.8M) like this:

Channel	Last 13 weeks	Next 13 weeks	Change	Why
Channel4	4.37M	**5.36M**	**+23%**	best return on the next unit of spend
Channel0	3.80M	**3.61M**	−5%	second-best marginal return
Channel2	1.44M	**1.23M**	−15%	
Channel1	3.20M	**2.24M**	**−30%**	weakest marginal return; cut to the limit
Channel3	7.95M	**5.56M**	**−30%**	biggest budget, saturated; cut to the limit
Total	20.77M	**18.00M**	−13%	

Expected revenue generated by this spend over the quarter (including the following weeks' carry-over): **23.0M (14.6M – 33.1M)**. Compared with cutting every channel by the same 13%, this mix adds **+0.3M (−1.1M to +1.8M)** — better in about 2 of 3 scenarios. The gain from re-mixing is modest and uncertain; the main decision — **spend less** — is the robust one.

If the budget changes:

Budget	Ch0	Ch1	Ch2	Ch3	Ch4	Revenue generated
24M	4.94M (+30%)	3.04M (−5%)	1.88M (+30%)	8.46M (+6%)	5.69M (+30%)	26.8M (17.4 – 38.4)
18M	3.61M	2.24M	1.23M	5.56M	5.36M	23.0M (14.6 – 33.1)
12M*	2.47M (−35%)	1.28M (−60%)	0.79M (−45%)	3.58M (−55%)	3.88M (−11%)	18.1M (10.9 – 27.1)

* 12M breaks the ±30% rule (the rule allows no less than 14.5M); shown with the rule relaxed. At the 14.5M floor (every channel −30%) revenue is 20.0M (12.9 – 29.3). Channel4 is protected at every budget; Channel3 and Channel1 absorb cuts first.

What each channel returns

Channel	Revenue per 1 spent, past year (ROI)	Revenue from the *next* 1 spent (marginal ROI)	Chance ROI > 1
Channel0	1.30 (0.29 – 2.97)	0.64 (0.12 – 1.46)	55%
Channel1	1.03 (0.31 – 2.27)	0.54 (0.14 – 1.17)	40%
Channel2	1.20 (0.46 – 2.21)	0.71 (0.20 – 1.41)	55%
Channel3	0.94 (0.38 – 1.71)	0.51 (0.20 – 0.95)	36%
Channel4	1.62 (0.36 – 3.35)	0.77 (0.16 – 1.47)	73%

Read it this way: on average each channel paid back about its cost in **revenue** over the past year, but at today's spend levels **the next unit of spend returns 0.5–0.8 in revenue in every channel**. Spending more buys less and less (at 24M the extra 6M would add only about 3.7M of revenue). These are revenue returns, not profit: with a gross margin below 100%, a channel needs an ROI well above 1 to make money — the CFO should apply the margin before deciding how much to cut.

The Channel3 experiment

The vendor reported that switching Channel3 off in 8 regions for 8 weeks showed a return of 0.8 (0.3 – 1.3). **Our data shows the switch-off did not happen:** Channel3 spend in those regions stayed at its normal level

(only 2 of 64 region-weeks had zero spend; their spend relative to other regions even rose slightly). We therefore did not let the reported number drive the budget. For what it's worth, our model's own estimate for the same regions and weeks is 1.10 (0.38 – 2.01) — consistent with the vendor's figure — and fully trusting the vendor would move the 18M plan by less than 0.4M in any channel. Before paying for more tests, we need to understand why this one was not executed as described.

How much to trust this

- **Checked on unseen weeks:** forecast of national weekly revenue for the 13 held-out weeks was off by 6.3% on average and ran 6% low (Q4 2023 was stronger than the model expected); 85% of weeks fell inside the 90% band.
- **Region-level beats national:** a model of national totals gives similar averages but wider ranges; most of the useful variation (65–79% of spend movement) is region-by-week, which national totals throw away.
- **The method was tested:** on simulated data with known answers, the model recovered the true returns inside its ranges 9 times out of 10.
- **Wide ranges are real, not a flaw:** channels move together and weekly sales are noisy; the data cannot pin returns down more tightly.

The next experiment — and why we should not run a small one

The uncertainty that costs the most money is in **Channel0 and Channel4** (each worth about 0.34M of better decisions this quarter if we knew it exactly). But region tests are weak here because weekly sales per region are very noisy: even switching Channel0 off in half the regions (20) for 16 weeks would detect a true return of ~1.3 only about **1 time in 3**, while giving up ~2M of spend. **Recommendation:** do not run a small test that cannot answer the question. Instead (1) move budget gradually in the direction above (one quarter at a time) and watch the results against this model; (2) if a test is run, make it large (≥ 20 regions, ≥ 16 weeks) and verify weekly that spend actually changed in the test regions.

What the model cannot tell you

Whether creative or targeting changes would shift these returns; effects beyond about two months; which **regions** to target (we kept each channel's regional mix); profit (we have revenue, not margin); and how much to trust the role we gave market sentiment. We treat it as something that drives both spend and sales; removing it from the model drops Channel3's estimate to about 0.3, so this matters. The alternative reading — advertising works partly *through* sentiment — would imply advertising lowers sales via sentiment, which is implausible, so we keep it. Anything that moved both spend and sales in the same region and week without being recorded would also bias these numbers.